

reponse_to_reviews

Reviewer 1

Referee report of “cardinalR: Generating Interesting High-Dimensional Data Structures”.

- Id: 2025-155

Recommendation (VT:: I copied this from the mail of the reviewer)

My overall impression from the package is positive. However, several things can be done to improve the paper and the package. Please see a brief summary below. More illustrations of using the code along with demo files should be provided. The authors should not oversell their package by saying that it can help evaluate clustering methods. If they insist on this package’s capability, perhaps, they need to outline how exactly they plan to achieve that. I agree that it can be helpful to simulate an interesting dataset, but its clustering complexity will be unknown. Grammar/professional presentation needs to be checked/improved. At places, the manuscript and package are slopy.

General

The manuscript “cardinalR: Generating Interesting High-Dimensional Data Structures” presents a wide variety of methods to simulate datasets of various shapes. While these capabilities are undoubtedly useful, the presentation of the material should be somewhat changed in my opinion. In particular, the authors need to provide more examples that would stimulate readers’ interest. Also, the authors should be careful with claiming that their package can be helpful for studying cluster analysis methods. To summarize, I think the package is useful, but presentation of the material should be improved.

Detailed comments

1. The authors claim that the package helps study clustering and dimension reduction algorithms. While I have no concerns with regard to dimension reduction algorithms, I would like to encourage the authors to provide more details regarding how the package can be used for studying cluster analysis procedures. This is very much related to Figure~1 provided by the authors. Having just shape generator routines is not sufficient to claim that the package is useful for evaluating clustering algorithms. The clustering complexity of a particular dataset is defined not just by the geometry of clusters themselves, but also by their proximity to and interaction with each other. For example, no matter what cluster shapes a user chooses, if the clusters are far enough from each other, even a simple k-means algorithm will find the perfect cluster assignment. In my opinion, the authors should say no more than that their package allows simulating clusters of different shapes. At the same time, this is not particularly useful for studying the systematic performance of clustering algorithms.
2. This point is closely related to the previous one. What is a cluster according to the authors’ definition? There are many alternative definitions. If the authors want to claim that their package can help evaluate

clustering algorithms based on datasets simulated from their package, there should be a discussion regarding what constitutes a cluster.

Please see some useful references below: - Hennig, C. (2015) What are the true clusters?, Pattern Recognition Letters, 6, 53-62. - Maitra, R. and Melnykov, V. (2010) Simulating data to study performance of finite mixture modeling and clustering algorithms, Journal of Computational and Graphical Statistics, 2(19), 354-376

3. page 2: “A full list of available shape generators are available...” - “IS available”; also, avoid using “available” twice
4. Don’t make space before “where” in the beginning of a line (see, for examples, page 6)
5. Please check the manuscript for English and typos.

Code

1. I think that the number of examples is insufficient. Many more code-related illustrations motivating the use of the package need to be provided, preferably with corresponding demo-files. These should not be massive, but can illustrate different aspects of the package use.
2. Please make sure that all help files and messages provided by the package are informative, professional, and grammatically correct; for example, remove “!!!” and correct the comments below “Data generation completed successfully!!! Multiple clusters generation completed successfully!!!”

Reviewer 2